

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): LU9915

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b001	1.4.2

Version History

Module Version	Date	Notes
1_0_0_0	5/1/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LU9915')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LU9915')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 8000, Model='LU9915')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command 3DFormat	Value 'Off' 'Top Bottom'	Value 'Auto' 'Frame Sequential'	Value 'Side by Side'
# 3DFormat example InterfaceName.Set('3DFormat', 'Off')			
Command 3DInvert	Value 'On'	Value 'Off'	
# 3DInvert example InterfaceName.Set('3DInvert', 'On')			
Command AspectRatio	Value '4:3' 'Auto' '5:4'	Value '16:9' 'Real' '1.88'	Value '16:10' 'Theaterscope' '2.35'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AutoImage	Value 'String'		
# AutoImage example InterfaceName.Set('AutoImage', 'String')			
Command DLPLink	Value 'On'	Value 'Off'	
# DLPLink example InterfaceName.Set('DLPLink', 'On')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer 1/YPbPr 1' 'HDMI' 'HDBaseT'	Value 'Computer 2/YPbPr 2' 'DisplayPort'	Value 'DVI-D' '3G-SDI'
# Input example InterfaceName.Set('Input', 'Computer 1/YPbPr 1')			

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Command LampMode	Value 'Normal'	Value 'Eco'	Value 'Custom'
# LampMode example InterfaceName.Set('LampMode', 'Normal')			
Command MenuNavigation	Value 'Menu On' 'Down' 'Enter'	Value 'Menu Off' 'Right'	Value 'Up' 'Left'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu On')			
Command PictureMode	Value 'Presentation' 'DICOM SIM'	Value 'Bright' 'Vivid'	Value 'Cinema'
# PictureMode example InterfaceName.Set('PictureMode', 'Presentation')			
Command PIPIInput	Value 'Computer 1/YPbPr 1' 'HDMI' 'HDBaseT'	Value 'Computer 2/YPbPr 2' 'DisplayPort'	Value 'DVI-D' '3G-SDI'
# PIPInput example InterfaceName.Set('PIPIInput', 'Computer 1/YPbPr 1')			
Command PIPMode	Value 'On'	Value 'Off'	
# PIPMode example InterfaceName.Set('PIPMode', 'On')			
Command PIPPosition	Value 'Top Left' 'Bottom Right'	Value 'Top Right' 'PBP'	Value 'Bottom Left'
# PIPPosition example InterfaceName.Set('PIPPosition', 'Top Left')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value '4:3' 'Auto' '5:4'	Value '16:9' 'Real' '1.88'	Value '16:10' 'Theaterscope' '2.35'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DLPLink	Value 'On'	Value 'Off'	
# DLPLink examples InterfaceName.Update('DLPLink') Value = InterfaceName.ReadStatus('DLPLink') InterfaceName.SubscribeStatus('DLPLink', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'Computer 1/YPbPr 1' 'HDMI'	Value 'Computer 2/YPbPr 2' 'DisplayPort'	Value 'DVI-D' '3G-SDI'

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	'HDBaseT'		
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Normal'	Value 'Eco'	Value 'Custom'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value —		
# LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command PictureMode	Value 'Presentation' 'DICOM SIM'	Value 'Bright' 'Vivid'	Value 'Cinema'
# PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command PIPIInput	Value 'Computer 1/YPbPr 1' 'HDMI' 'HDBaseT'	Value 'Computer 2/YPbPr 2' 'DisplayPort'	Value 'DVI-D' '3G-SDI'
# PIPInput examples InterfaceName.Update('PIPIInput') Value = InterfaceName.ReadStatus('PIPIInput') InterfaceName.SubscribeStatus('PIPIInput', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

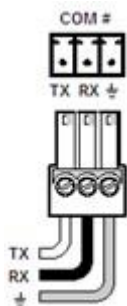
Data Bits: 8

Parity: None

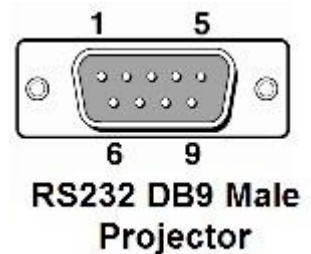
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	8000
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

3D Format Auto	\x0D*3d=auto#\x0D
3D Format Frame Sequential	\x0D*3d=fs#\x0D
3D Format Off	\x0D*3d=off#\x0D
3D Format Side by Side	\x0D*3d=sbs#\x0D
3D Format Top Bottom	\x0D*3d=tb#\x0D
3D Invert Off	\x0D*3d=da#\x0D
3D Invert On	\x0D*3d=iv#\x0D
Aspect Ratio 1.88	\x0D*asp=1.88#\x0D
Aspect Ratio 16:10	\x0D*asp=16:10#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 2.35	\x0D*asp=2.35#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio 5:4	\x0D*asp=5:4#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Real	\x0D*asp=REAL#\x0D
Aspect Ratio Theaterscope	\x0D*asp=THEA#\x0D
Auto Image None	\x0D*auto#\x0D
DLP Link Off	\x0D*dlplink=off#\x0D
DLP Link On	\x0D*dlplink=on#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input 3G-SDI	\x0D*sour=sdi#\x0D
Input Computer 1/YPbPr 1	\x0D*sour=RGB#\x0D
Input Computer 2/YPbPr 2	\x0D*sour=RGB2#\x0D
Input DVI-D	\x0D*sour=dvid#\x0D
Input DisplayPort	\x0D*sour=dp#\x0D
Input HDBaseT	\x0D*sour=hdbaset#\x0D
Input HDMI	\x0D*sour=hdm#\x0D
Lamp Mode Custom	\x0D*lampm=cust#\x0D
Lamp Mode Eco	\x0D*lampm=eco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Menu Off	\x0D*menu=off#\x0D
Menu Navigation Menu On	\x0D*menu=on#\x0D

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Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D
PIP Input 3G-SDI	\x0D*psour=sdi#\x0D
PIP Input Computer 1/YPbPr 1	\x0D*psour=RGB\x0D
PIP Input Computer 2/YPbPr 2	\x0D*psour=RGB2\x0D
PIP Input DVI-D	\x0D*psour=dvid#\x0D
PIP Input DisplayPort	\x0D*psour=dp#\x0D
PIP Input HDBaseT	\x0D*psour=hdbaset#\x0D
PIP Input HDMI	\x0D*psour=hmi#\x0D
PIP Mode Off	\x0D*pip=off#\x0D
PIP Mode On	\x0D*pip=on#\x0D
PIP Position Bottom Left	\x0D*pippos=bl#\x0D
PIP Position Bottom Right	\x0D*pippos=br#\x0D
PIP Position PBP	\x0D*pippos=pbp#\x0D
PIP Position Top Left	\x0D*pippos=tl#\x0D
PIP Position Top Right	\x0D*pippos=tr#\x0D
Picture Mode Bright	\x0D*appmod=bright#\x0D
Picture Mode Cinema	\x0D*appmod=cine#\x0D
Picture Mode DICOM SIM	\x0D*appmod=dicom#\x0D
Picture Mode Presentation	\x0D*appmod=preset#\x0D
Picture Mode Vivid	\x0D*appmod=vivid#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D

Appendix B. Update Commands

Aspect Ratio	\x0D*asp=?#\x0D
DLP Link	\x0D*dplink=?#\x0D
Freeze	\x0D*freeze=?#\x0D
Input	\x0D*sour=?#\x0D
Lamp Mode	\x0D*lampm=?#\x0D
Lamp Usage	\x0D*ltim=?#\x0D
Picture Mode	\x0D*appmod=?#\x0D
PIP Input	\x0D*psour=?#\x0D
Power	\x0D*pow=?#\x0D
Video Mute	\x0D*blank=?#\x0D